

Brijodison

Engineering Notebook Element K

Engineering Service Learning

Reflection on the Design Project

The first step in our design process was assigning group roles in order to hold each other accountable. This step was important because we were able to get to know each other before we worked on this project. After we finished that, we identified the problem as well as collected data and measurements; these include Mr. Kiser's height, wingspan, and also the measurements of the cart. Mr. Kiser also gave us criteria and constraints for this project. An example of our criteria would be him not having to bend down to attach and detach our device. An example of constraint was our \$30 budget.

The next major step was proposing possible solutions without considering the ramifications of each design. This step was necessary in order to think outside the box and potentially put one of the proposed ideas into action. We came up with a total of almost 30 ideas, some of which were crazy and nearly impossible. After this, we narrowed our designs down to only 6 of them. We came together collectively and decided which 6 would be most feasible based on our criteria as well as constraints. After we agreed on 6 proposed designs, we made a decision matrix to determine which design would be best fit.

Our following step was putting our chosen design into action. We began building our first prototype, but even after the matrix we decided to adapt a lot as we were constructing it. The prototype we ended up with was two PVC poles that go inside the front handle on the cart and attach to the bottom of the cart. This prototype was clearly the best option in regards to our

identified criteria and constraints. Even though we felt good about this, there were still a few problems such as too much sliding for our poles and structural issues.

The succeeding step was getting an outside opinion. This step came into fruition when several professionals came in on a day we called the “Designathon”. This day was important and led us to making some of the major adjustments to our prototype to get to our current product. Our first outside opinion was from Jared Lawson, a graduate student commencing research over design, modeling, and control of surgical robots at Vanderbilt. He proposed an idea that would connect our two poles in order to increase stability. While we didn’t do that exactly, we still decided to make sure our poles were tied together when being stored in order to solve the problem of potentially losing one of them. Our second outside opinion was from Peter Hermann, a member of the sales force. He had suggested crossing our poles in order to decrease their sliding along the base of the cart. While this idea seemed great at first, we would have had to implement hinges on the actual connectors to the cart. We came up with an alternative solution with the implementation of rubber along the base to ensure max security. Not only was this useful along the base (1st point of contact), this still allowed the poles to lay along the handle of the cart. We also later put rubber around the second and third point of contact after it still slid out of place. These rubber strips made the poles stay almost entirely still.

Our last step was tweaking our final prototype. This included shaping the PVC poles out to fit even better as well as finish off the glue where the PVC poles met the handles attached at the base of the cart. We had taken all professional opinions into account in order to develop our best prototype and while it isn’t perfect yet, it fits the criteria and constraints we were given at maximum efficiency while still solving the problem.

During our process of designing and creating our prototype we had the benefit of learning many lessons that we could apply to this project and more to come. We not only learned about information connected to this project, but advice we could use throughout life. In the beginning of our project we had a little bit of a communication problem where we didn't know how to convey our ideas to each other, however, through thoroughly talking it out we were able to find a way to understand each other's strengths and weaknesses. During our designing process we were able to each use our unique skills to collaborate with each other effectively. This entire project has benefited our knowledge in many different ways.